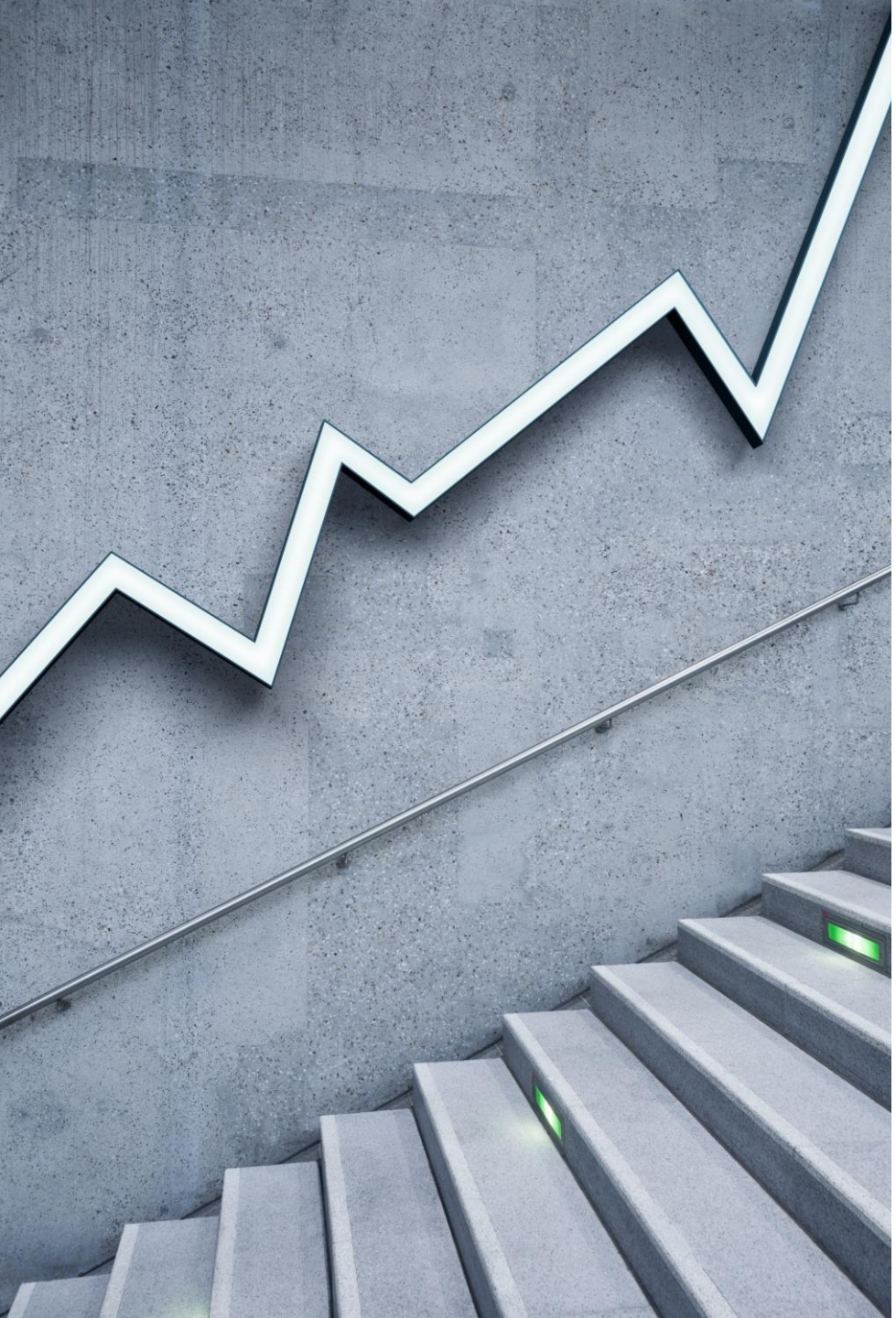


Social Network Analysis

NETWORK MEASURES

SLIDE CREDITS: TEACHING MATERIAL ON SOCIAL NETWORK ANALYSIS BY TANMOY CHAKRABORTY, WILEY, 2021

Network Measures: Classification

- Microscopic Level
  - We begin by analyzing how a pair of nodes interacts and gradually trace the interactions at the group level or subgraph level.
  - *Dyadic level* ➔ Interaction patterns among two nodes
  - *Triadic level* ➔ Interaction patterns among three nodes
  - *Ego-centric circles* ➔ Interaction pattern between ego node with its alters
  - Network Measures for Microscopic Level Analysis
    - *Degree*
    - *Local clustering coefficient*
    - *Node centrality*

Network Measures: Classification

- **Mesoscopic**
  - Mesoscopic analysis is an intermediary between microscopic and macroscopic analyses, which mostly deals with a subset of the entire population.
  - ***Communities*** ➔ Formed due to frequent interactions among homogeneous nodes in a network
    - Within a community, the nodes exhibit a particular kind of dynamicity
    - Across communities, the dynamic behaviour differs
  - ***Network Motifs*** ➔ Subgraphs that repeat themselves frequently within or across a network
    - Highly effective in capturing functional properties in a network
- **Network Measures for Microscopic Level Analysis**
  - Connected components
  - Giant components
  - Group centralities

Network Measures: Classification

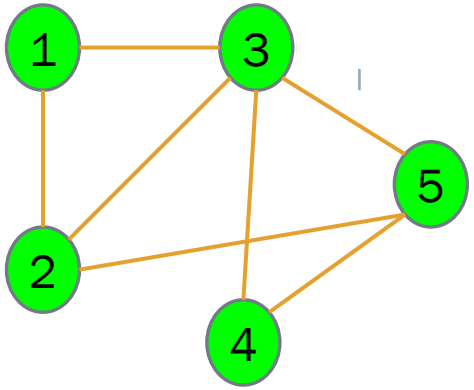
- Macroscopic

- At macroscopic level, we deal with the entire network as a whole and try to understand the micro-level dynamics by exploring the overall graph property.

- Features of Interest :
  - Connectedness,
  - Diameter or Average path length,
  - Degree Distribution,
  - Edge Density, etc.

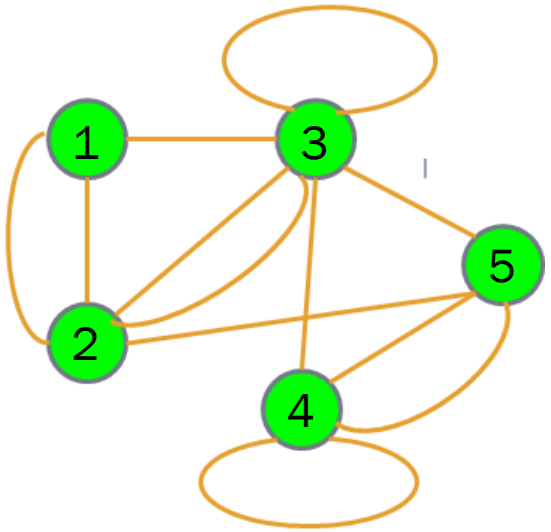
- Network Measures for Microscopic Level Analysis
  - Degree Distribution
  - Path and Diameter
  - Edge density
  - Global clustering coefficient
  - Reciprocity and Assortativity

Degree of a Node



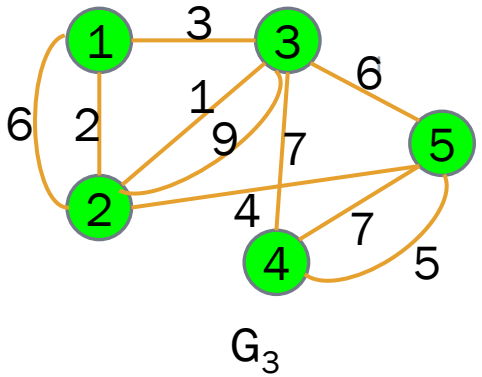
G_1

- For an undirected, unweighted network, degree of a node ***v*** is defined as the number of nodes in the network to which there is an edge from the node ***v***.
- In other words, for an undirected, unweighted network, degree of a node ***v*** is the number of edges of the network that are incident on the node ***v***.
- Putting differently, for an undirected, unweighted network, degree of a node ***v*** is the number of neighbours of the node ***v***.
- In graph G_1, degrees of the nodes 1 through 5 are 2, 3, 4, 2, 3.
- In graph G_2, degrees of the nodes 1 through 5 are 3, 5, 7, 5, 4.
- Note: A self-loop is counted twice in evaluating degree of a node.



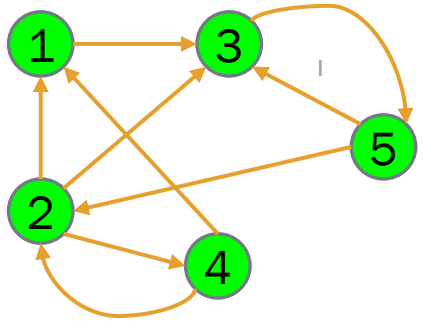
G_2

Weighted Degree of a Node



- For an undirected, weighted network, the weighted degree of a node is defined as the sum of weights of the edges incidents on that node
- For the weighted undirected graph G_3, the weighted degrees of the nodes are as follows:
  - Weighted degree of node 1 is 11
  - Weighted degree of node 2 is 22
  - Weighted degree of node 3 is 26
  - Weighted degree of node 4 is 16
  - Weighted degree of node 5 is 22

Indegree and Outdegree of a Node



G_4

- In a directed network, the indegree of a node is defined as the number of incoming edges to the node
- In a directed network, the outdegree of a node is defined as the number of outgoing edges from the node
- For the directed graph G_4, the indegrees and outdegrees of the nodes are as follows:
  - Indegrees of the nodes 1 through 5 are 2, 2, 3, 1, 1
  - Outdegrees of the nodes 1 through 5 are 1, 3, 1, 2, 2

Sum of the Degrees…

- For an unweighted, undirected network, the sum of the degrees of the nodes in a graph is twice the number of edges in the graph

- Proof
  - ✓ When we add an edge **e** to graph, it joins a pair of vertices v_i and v_j of the graph.
  - ✓ Prior to the addition of the edge **e** to graph, let the degrees of the nodes v_i and v_j be d_i and d_j .
  - ✓ After addition of the edge **e** to graph, the revised degrees of the nodes v_i and v_j be $d_i + 1$ and $d_j + 1$. The degrees of the other nodes remain unaffected.
  - ✓ Then, on addition of an edge **e**, the sum of degrees of the nodes in G is incremented by 2 from its previous value. The fact is true for the addition of any edge to the graph.
  - ✓ If we add $|E|$ number of edges to the graph one-by-one, the sum of the degrees is enhanced by $2 \times |E|$.
  - ✓ If a graph has no edges, all the nodes have degree zero, and so, the sum of the degrees is zero.
  - ✓ Thus, a graph with $|E|$ edges has its sum of the degrees of the nodes as $2 \times |E|$.

Sum of the Weighted Degrees…

- Sum of the weighted degrees of the nodes in an undirected weighted graph is twice the sum of weights of the edges in the graph
  - Proof: Proved following the same line of approach

Sum of Indegrees and Outdegrees

- In a directed network, the sum of indegrees is same as the sum of outdegrees.
- Proof. Proved following the same line of approach

Number of Odd-degree Nodes…

- Number of odd-degree nodes in an undirected network is always even.

- Proof.
  - ✓ If possible, let the number of odd degree nodes of the graph $G(V, E)$ be an odd integer.
  - ✓ Then, the sum of the degrees of these odd-degree nodes is an odd integer, say N_{odd}.
  - ✓ All the remaining nodes of the graph have even degrees.
  - ✓ Clearly, the sum of the degrees of these even-degree nodes is an even integer, say N_{even}.
  - ✓ Then, the sum of the degrees of all the nodes of the graph is $N_{odd} + N_{even}$, which is odd integer.
  - ✓ However, the sum of the degrees of all the nodes is $2 \times |E|$
  - ✓ So, $N_{odd} + N_{even} = 2 \times |E|$, which is a contradiction, as the LHS is odd and RHS is even!
  - ✓ Hence the result.

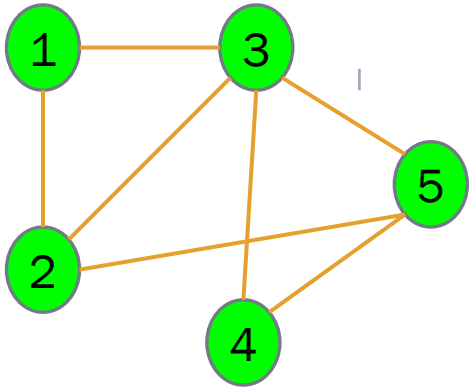
Degree Distribution

- Degree distribution of a network is the (probability) distribution of the degrees of nodes over the whole network.
- Let a network has $N = |V|$ nodes.
- Let P_k denotes the probability that a randomly chosen node has degree k.
- Then, $P_k = \frac{N_k}{N}$, where N_k refers to the number of nodes of degree k in the network.
- The distribution (k, P_k) represents the degree distribution of the concerned graph,
- The mean degree, denoted $\langle k \rangle$, is given by $\langle k \rangle = \sum_k k.P_k$.

Cumulative Degree Distribution

- Cumulative degree distribution (CDD) is given by the fraction of nodes with degree smaller than or equal to k.
- In other words, it is the distribution(k, C_k), where $C_k = \frac{\sum_{k' \leq k} N_{k'}}{N}$
- Complementary cumulative degree distribution (CCDD) is given by the fraction of nodes with degree greater than or equal to k.
- In other words, it is the distribution(k, CC_k), where $CC_k = 1 - C_k$

Degree Distribution: Example

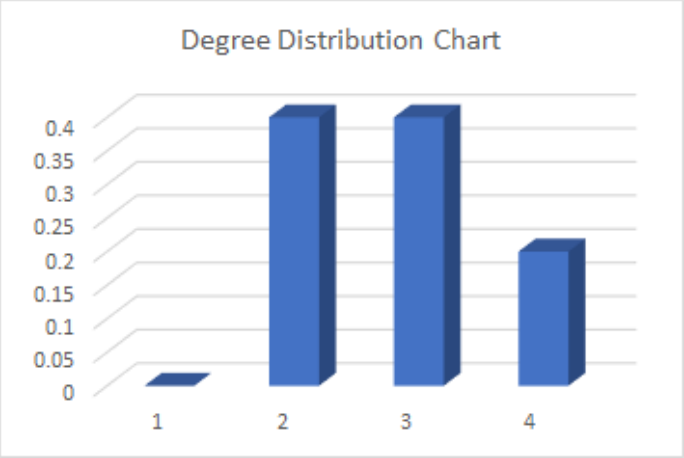


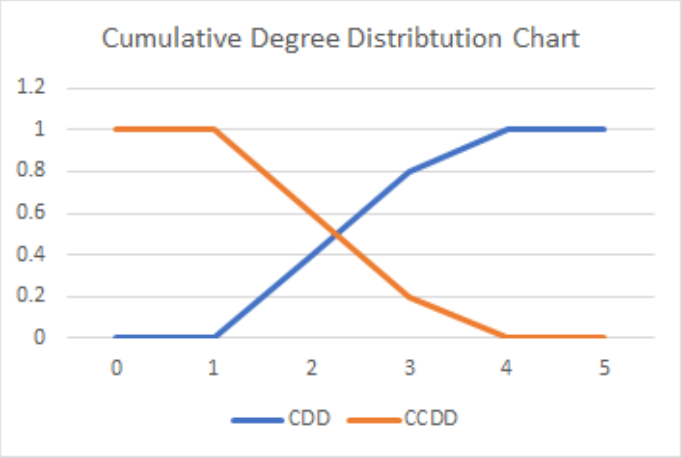
G_1

- For the graph G_1, we have the following:
  $N = 5$, and $N_1 = 0, N_2 = 2, N_3 = 2, N_4 = 1.$

- The above implies, $P_1 = 0, P_2 = 0.4, P_3 = 0.4, P_4 = 0.2,$
  $C_1 = 0, C_2 = 0.4, C_3 = 0.8, C_4 = 1.0,$
  and $CC_1 = 1.0, CC_2 = 0.6, CC_3 = 0.2\ CC_4 = 0.0.$

- Then the degree distribution, Cumulative degree distribution and Complementary cumulative degree distribution are as follows:





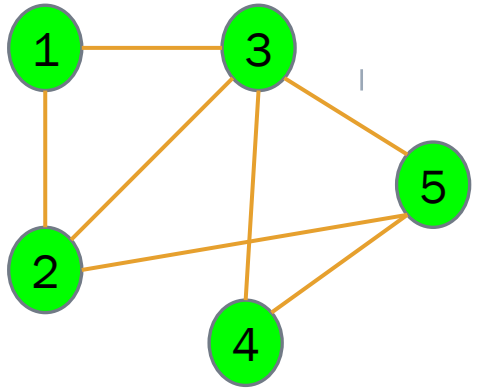
Some Graph Preliminaries…

- The density of a graph $G(V, E)$, denoted $\rho(G)$, is defined as the ratio of the number of edges in the graph to the total number of possible edges in the network. Mathematically,

$$\rho(G) = \frac{2 \times |E|}{|V| \times (|V| - 1)}$$

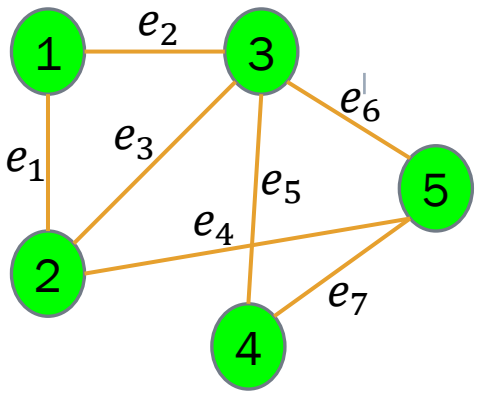
- For the graph G_1, the network density is:

$$\frac{2 \times 7}{5 \times 4} = 0.7$$



G_1

Some Graph Preliminaries…



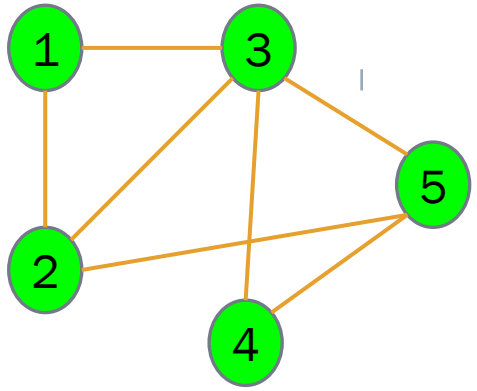
G_1

- The distance between nodes v_i and v_j in a graph is defined as the length of the shortest path between the nodes v_i and v_j.
- In graph G_1, the distance between 1 and 4 is 2, the same between 1 and 5 is also 2.
- The diameter of a network is defined as the maximum distance between any pair of nodes in the network.
- The diameter of the graph G_1 is 2.

Some Graph Preliminaries…

- For a graph G with n nodes, the average path length l_G is defined as the average number of steps along the shortest paths for all possible pairs of nodes in the network.

$$l_G = \frac{\sum_{i \neq j} d_{ij}}{n(n-1)}, \text{ where } d_{ij} \text{ is distance between nodes } v_i \text{ and } v_j$$

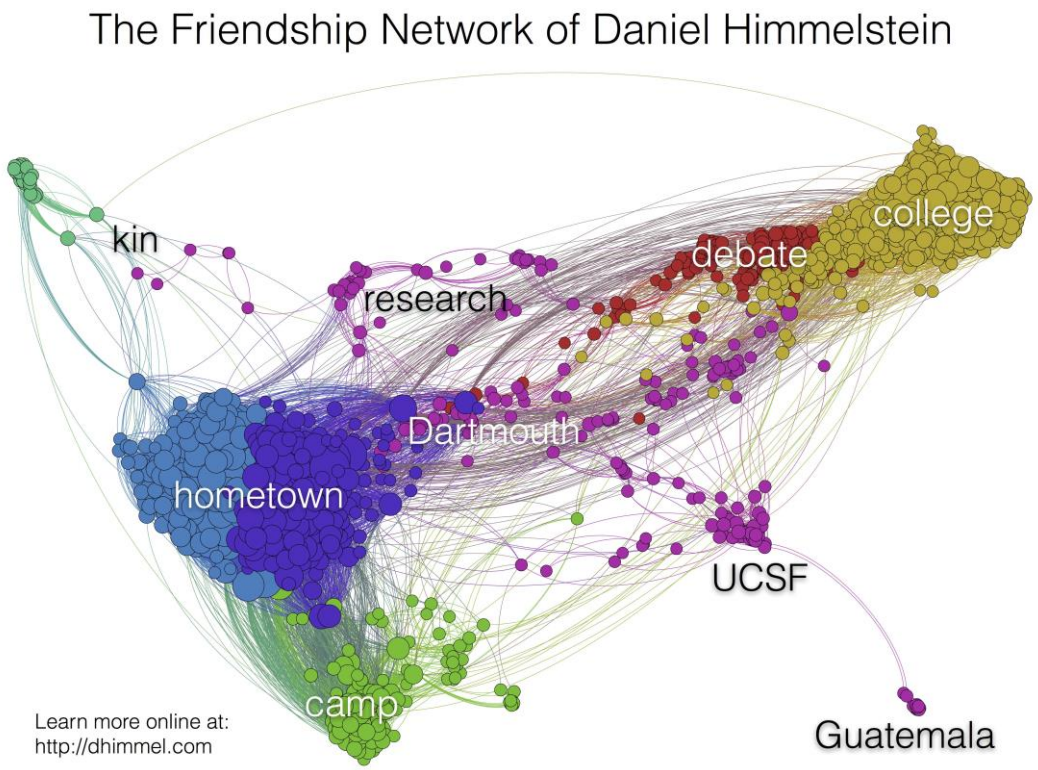


G_1

- For the graph G_1, the average path length is:

$$\frac{2 \times (1+1+2+2+1+2+1+1+1+1)}{5 \times 4} = \frac{26}{20} = 1.3$$

Clusters in Social Networks



A Facebook Friendship Network Example

https://blog.dhimmel.com/friendship-network/

- In social networks, we often find
  - tightly-knit groups here and there
  - less dense ties away from these groups
- Indicative of friendship structures in social media
- Measure used to capture these phenomena
  - Local clustering coefficient
  - Global clustering coefficient

Local Clustering Coefficient

The local clustering coefficient C_i for a vertex v_i in a network $G(V, E)$ is given by the proportion of edges between the vertices within its neighborhood divided by the number of links that could possibly exist between them.

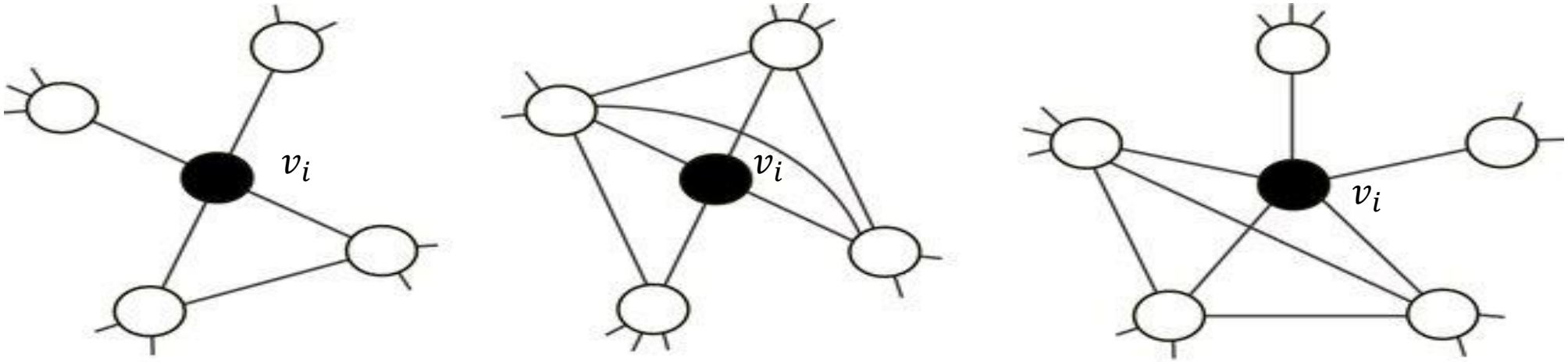
$$C_i = \frac{2 \times |\{e_{jk} | \, v_j, v_k \in N_i, e_{jk} \in E\}|}{k_i.(k_i - 1)}$$

Where N_i is the neighbourhood of the vertex v_i, and $k_i = |N_i|$.

In other words, in a network $G(V, E)$, the local clustering coefficient of node $v_i \in V$, denoted C_i, is defined as

$$C_i = \frac{Number\ of\ edges\ between\ neighbors\ of\ v_i}{Number\ of\ maximum\ possible\ edges\ between\ neighbors\ of\ v_i}$$

Local Clustering Coefficient

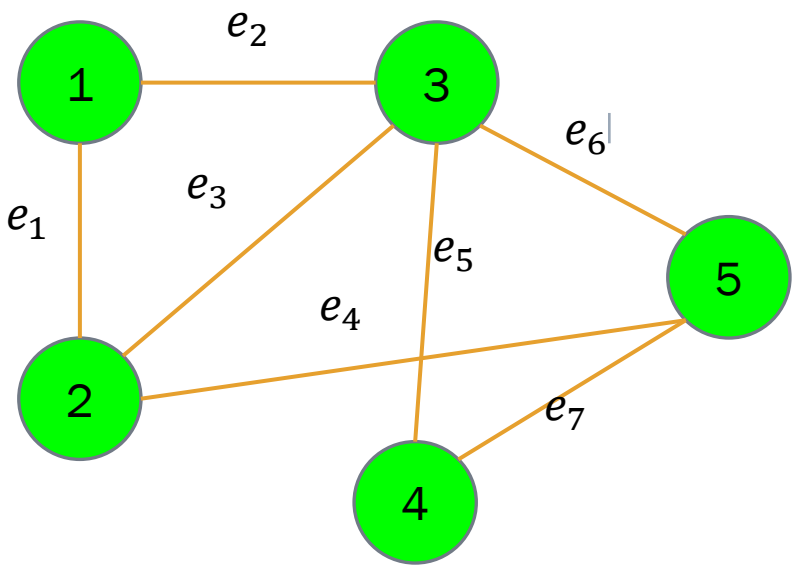


$k_i = 4$ $\quad$ $k_i = 4$ $\quad$ $k_i = 5$

$C_i = \frac{1}{6}$ $\quad$ $C_i = \frac{2}{3}$ $\quad$ $C_i = \frac{3}{10}$

https://www.researchgate.net/publication/236604411_Suicide_Ideation_of_Individuals_in_Online_Social_Networks/figures?lo=1

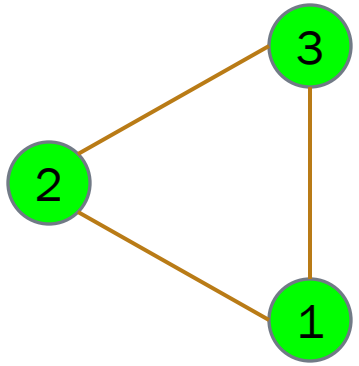
Local Clustering Coefficient



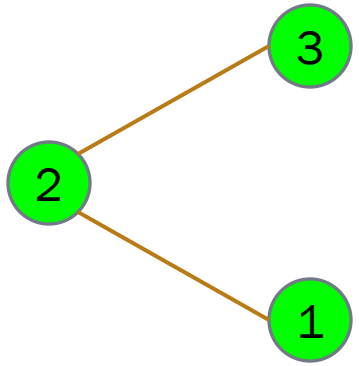
G_1

- In graph G_1,
  - the local clustering coefficient of node 2 is $^2/_3$
  - the local clustering coefficient of node 3 is $^3/_6$ i.e. $^1/_2$
  - and so on…

Global Clustering Coefficient



Closed Triplet

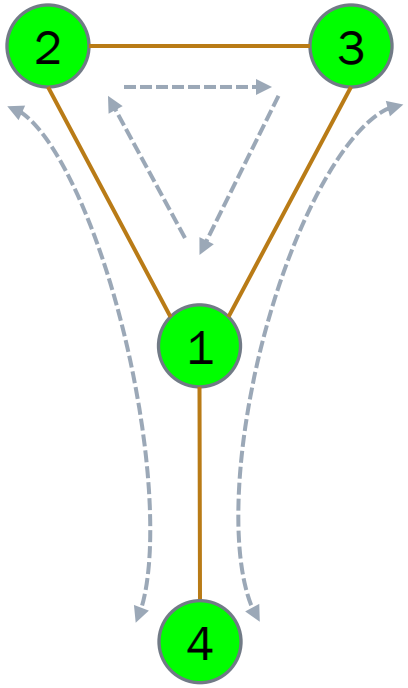


Open Triplet

- The global clustering coefficient C of a network G is defined as

$$C = \frac{Total\ number\ of\ closed\ triplets\ in\ G}{Total\ number\ of\ triplets\ (open\ \&\ closed)\ in\ G}$$

- In the graph G_2 , there is three closed triplet viz., [1,2,3], [2,3,1], and [3,1,2].
- In the graph G_2 , there are five open and closed triplets, viz., (1,2,3), (2,3,1), (3,1,2), (2,1,4), and (3,1,4).
- Thus, the global clustering coefficient of the graph G_2 is $^3/_5$.



G_2

Global Clustering Coefficient

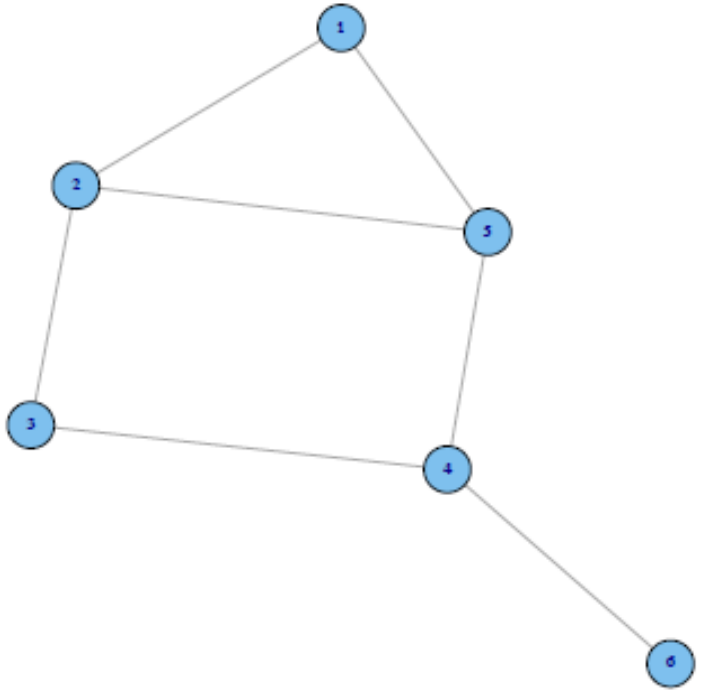
- The global clustering coefficient may also be written as

$$C = \frac{3 \times Total\ number\ of\ triangles\ in\ G}{Total\ number\ of\ triplets\ (open\ \&\ closed)\ in\ G}$$

- In other words, if $A = (A_{ij})$ is the adjacency matrix of the graph G, then

$$C = \frac{\sum_{i,j,k}(A_{ij}A_{jk}A_{ki})}{\sum_i k_i(k_i - 1)} \text{ where } k_i = \sum_j A_{ij}$$

Global Clustering Coefficient



1) Degree distribution

2) Average Degree

3) Average path length

4) Density

5) Local Clustering coefficient

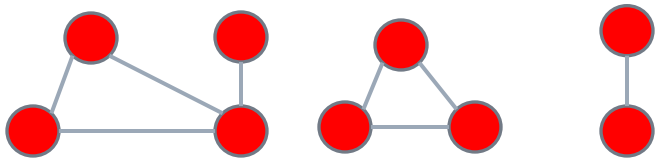
6) Global Clustering Coefficient

Connected Components

- In a typical social network, there are loose links that connects the tightly-knit clusters
- In an undirected network G, two nodes v_i and v_j are said to be connected if there exists a path between v_i and v_j .
- An entire network is said to be connected if any pair of nodes in the network is connected.
- Connected subnetworks of a network, if exist, are called components of the network.
- In real-world networks, there often exist one giant component (consuming major chunk of nodes) and many smaller components.
- In a network, connectedness shows resilience to link breakdowns.

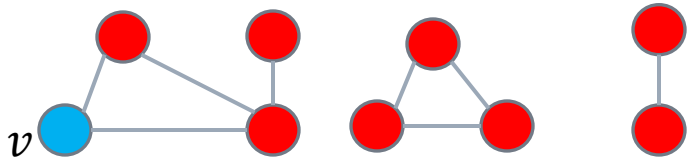
Finding Connected Components

$n = 0$



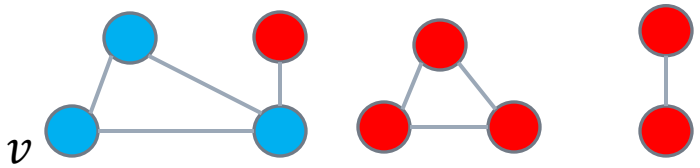
The network G with all nodes coloured red

$n = 1$



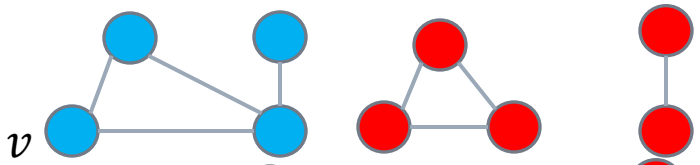
Choose a random node v, colour it blue, and set n to 1

$n = 3$



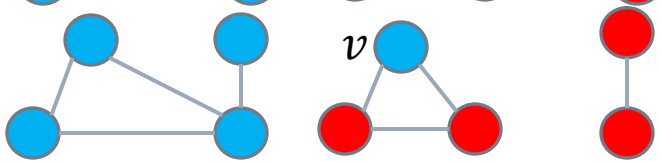
Apply BFS from node v, and colour with blue all the nodes reached thereof, and increment n each time

$n = 4$



No more node can be reached from v using BFS. We get a component in blue.

$n = 4$

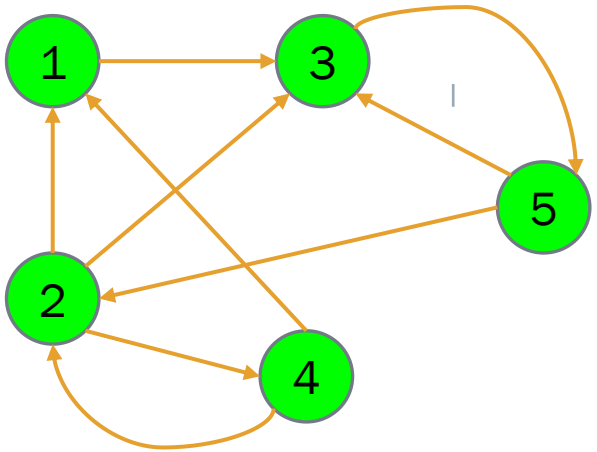


Since $n \neq 9$, we choose a red node as v, repeat the steps above to find other components

Connectedness in Directed Networks

- A directed network G is strongly connected if there exists a (directed) path between every pair of nodes in G.
- If we replace all the directed edges of a directed network G with undirected edges, then the resultant network is called an undirected version of the directed network G.
- A directed network G is said to be weakly connected if its undirected version is connected.

Can you say the below graph G_4 is strongly connected or weakly connected?



G_4